CS203 | FAST & FAIR GRADING KEY

Data Science Fundamentals - Midterm Fall 2026 | INSTRUCTOR ONLY | 100 points

PART 1 - MULTIPLE CHOICE

2 marks per correct answer; otherwise 0 | no negative marking

1	2	3	4	5	6	7	8	9	10
D	**C**	**A**	**A**	**C**	**D**	**D**	**B**	**C**	**B**
11	12	13	14	15	16	17	18	19	20
D	**B**	**A**	**B**	**A**	**A**	**C**	**D**	**B**	**C**

PART 2 - WRITTEN (6 x 10 = 60)

Award only earned points in each subpart

Q21 Counting bikes / 10

a) **1** fleet bike is the unit; **1** B17 and B24 serviced; **1** fraction = **2/3**. 3p

b) **1** B31 missing from matched join; **1** B17 has 2 jobs but is 1 bike; **1** use all 3 fleet bikes / distinct IDs; **1** B31 unserviced and B17 counted once. 4p

c) **1+1** any two excluded groups (other stations/times/weekends); **1** sample across stations, times and days. 3p

Q22 Means & spread / 10

a) **1** East mean **4**; **1** median **4**; **1** range **6**. 3p

b) **1** same centre ≠ same behaviour; **1** East range 6 vs West 2; **1** West more consistent here; **1** do not generalize beyond these logs. 4p

c) **1** added zero lowers mean; **1** range increases; **1** new range **8** (was 6). 3p

Q23 Data cleaning / 10

a) **1** remove extra T02; **1** 0.5 hour → **30 min**; **1** −4 invalid / unknown; **1** keep verified **240**. 4p

b) **1** cannot give true mean of all 5; **1** four valid values give **74.25**, one unknown; **1** label as observed-only mean. 3p

c) **1** valid-value median **22.5**; **1** imputation is estimate, not measurement; **1** flag/report method and uncertainty. 3p

Q24 Preprocessing / 10

a) **1** all-row median **40**; **1** train-only median **30**; **1** fit on train, reuse for test; **1** all-row fit leaks test information. 4p

b) **1** numeric station code adds fake rank/distance; **1** one-hot (or valid nominal) encoding; **1** preserve category identity, not numeric magnitude. 3p

c) **1** later fault confirmation = future leakage; **1** cycle count or station is valid; **1** use information known at dispatch. 3p

Q25 Correlation & pairing / 10

a) Rental count 10,20,30,40: **2** positive pairs with repair times 20,40,60,80; **2** negative pairs with 80,60,40,20. Equivalent valid pairs OK. 4p

b) **1** same marginal lists can have opposite association; **1** date-by-date pairs are missing; **1** recover paired daily records. 3p

c) **1** correlation does not prove causation; **1** plausible factor tied to both; **1** compare within groups / controlled study. 3p

Q26 Group comparisons / 10

a) **0.5 each** low-use East **10%**, West **15%**; high-use East **60%**, West **70%**; **1** East lower in both. 3p

b) **1** pooled East **40%**; **1** West **26%**; **1** high-use mix: East **60%**, West **20%**; **1** group mix reverses ranking. 4p

c) **1** East lower with 50:50 mix (35% vs 42.5%); **1** same weights applied to both; **1** no causal claim from observational data. 3p

FAST FAIR RULE

Give each listed point separately. Accept equivalent reasoning; do not penalize a carried-forward arithmetic slip twice. Ignore minor language errors. If an answer is borderline, apply the same rule to every paper.

SCORE: MCQs _____/40 Q21 _____ Q22 _____ Q23 _____ Q24 _____ Q25 _____ Q26 _____

TOTAL _____ / 100

Student ID: _____